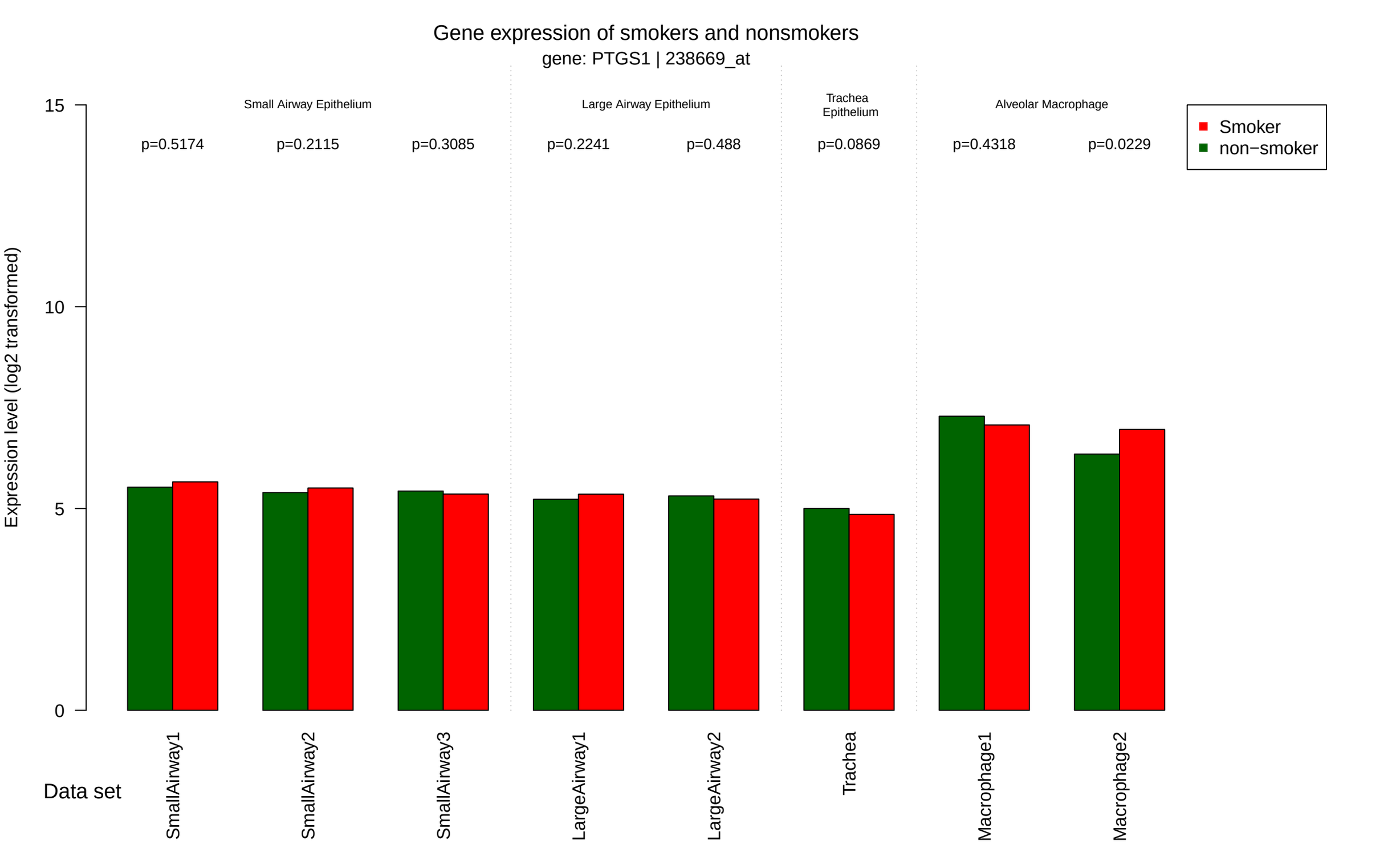
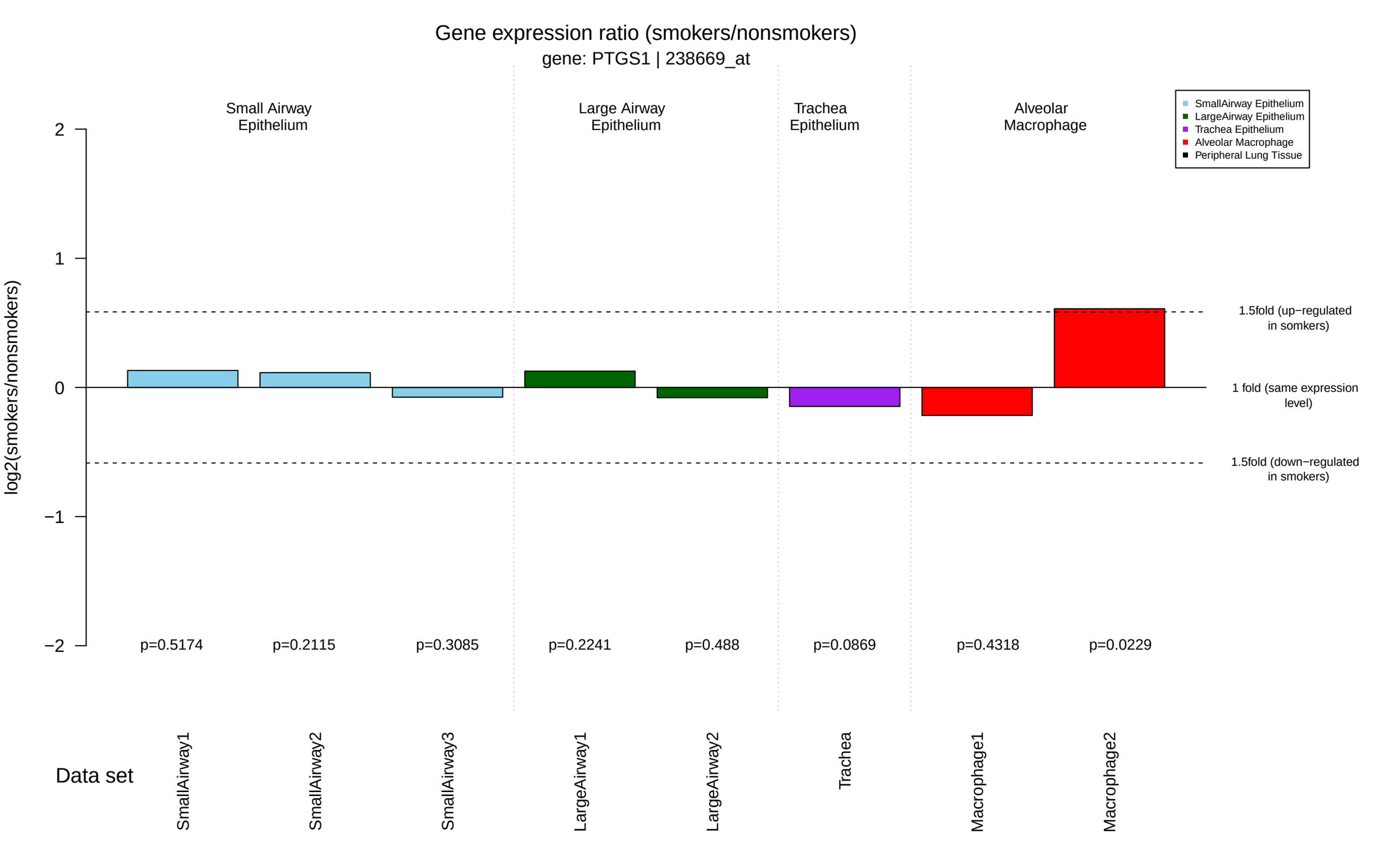
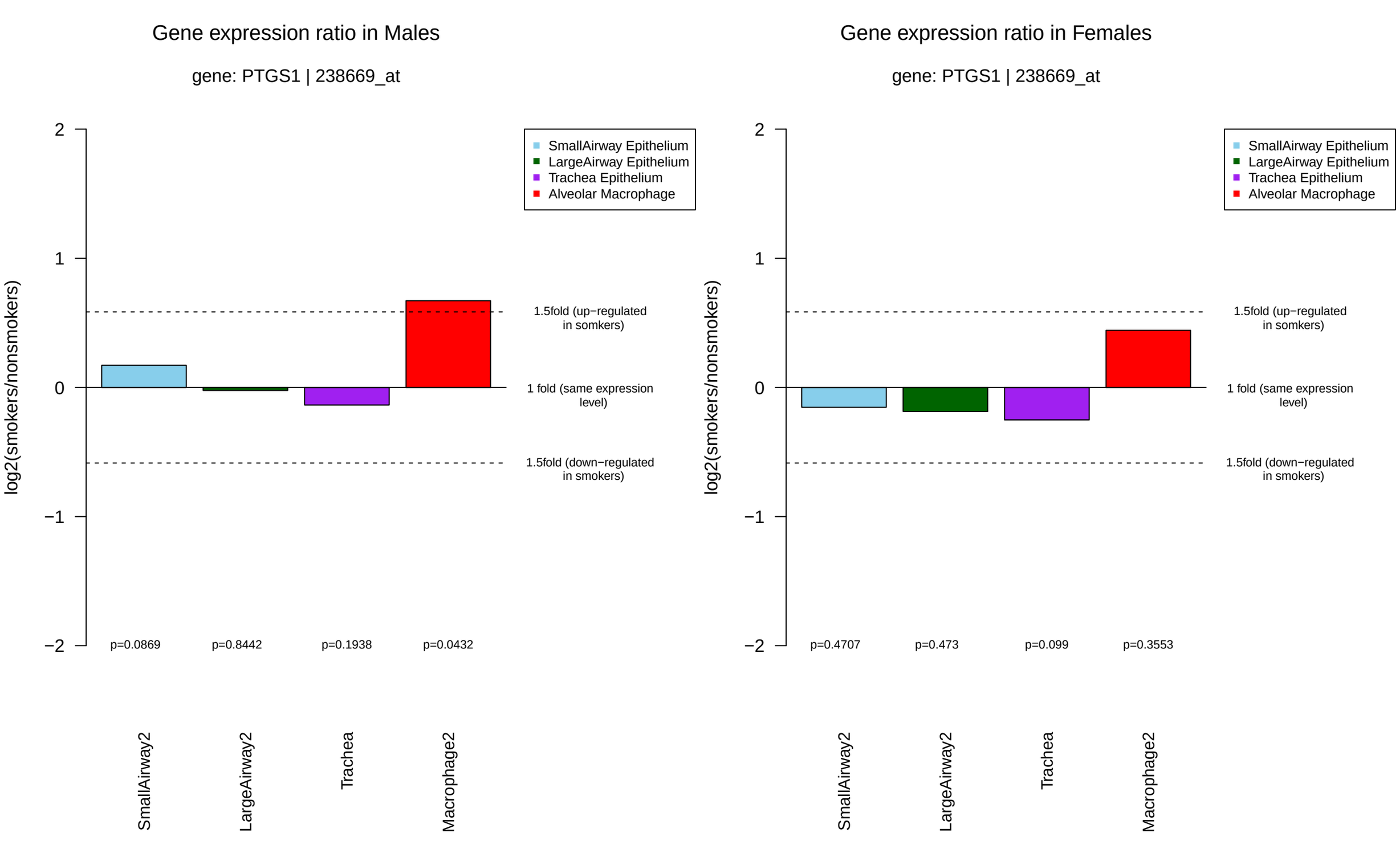
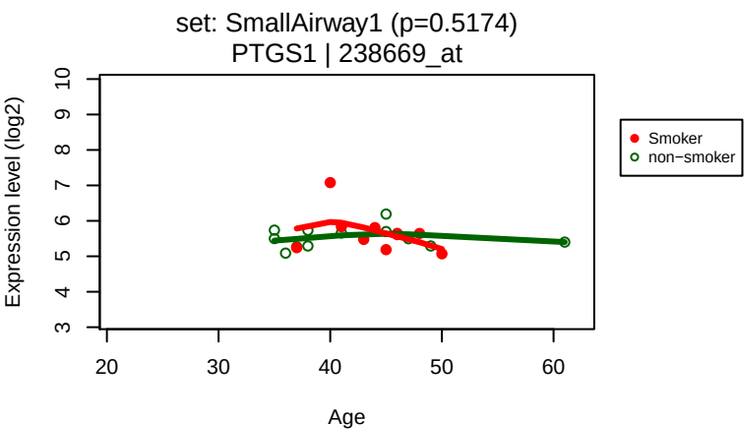
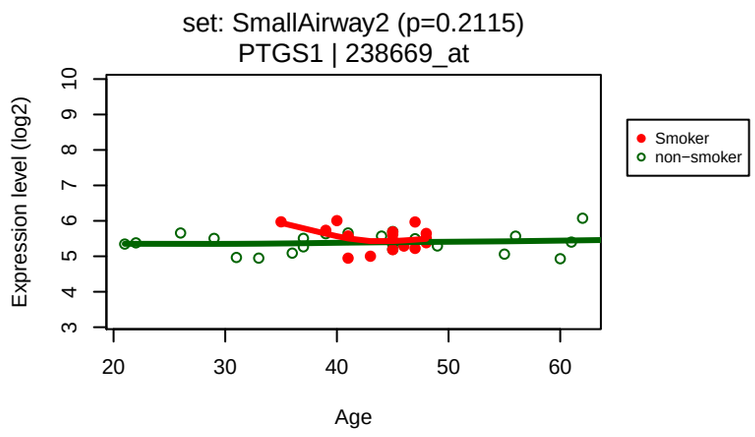
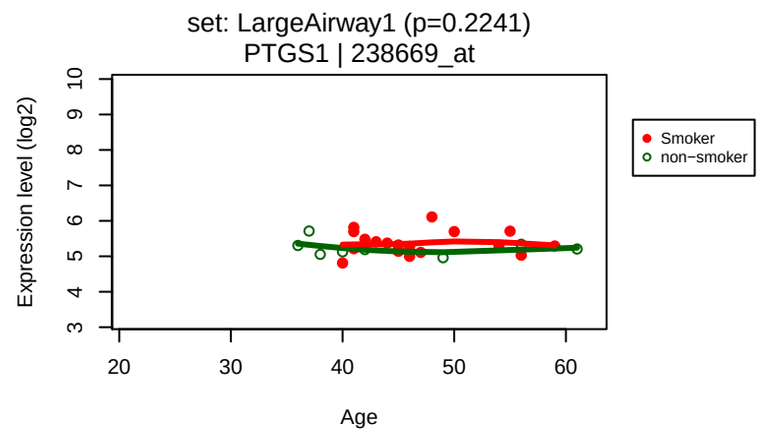

Gene expression of smokers and nonsmokers
gene: PTGS1 | 238669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5174
p=0.2115
p=0.3085
p=0.2241
p=0.488
p=0.0869
p=0.4318
p=0.0229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

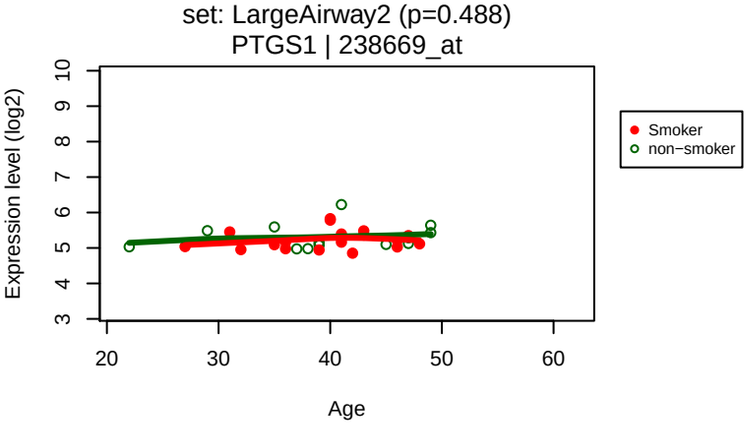

Gene expression ratio (smokers/nonsmokers)
gene: PTGS1 | 238669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5174
p=0.2115
p=0.3085
p=0.2241
p=0.488
p=0.0869
p=0.4318
p=0.0229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

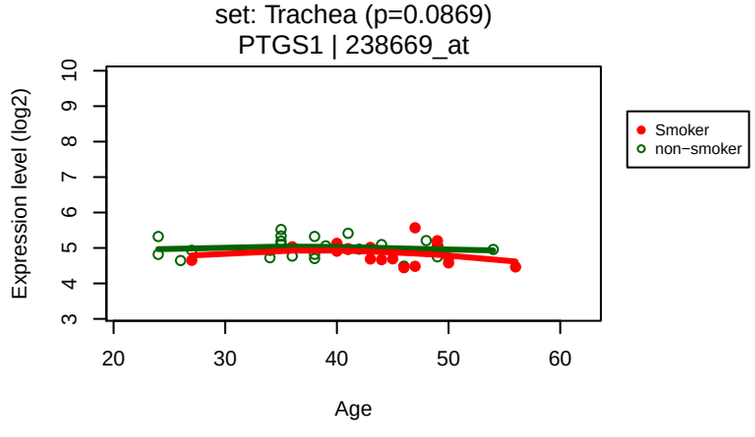

Gene expression ratio in Males
gene: PTGS1 | 238669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0869
p=0.8442
p=0.1938
p=0.0432
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGS1 | 238669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4707
p=0.473
p=0.099
p=0.3553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

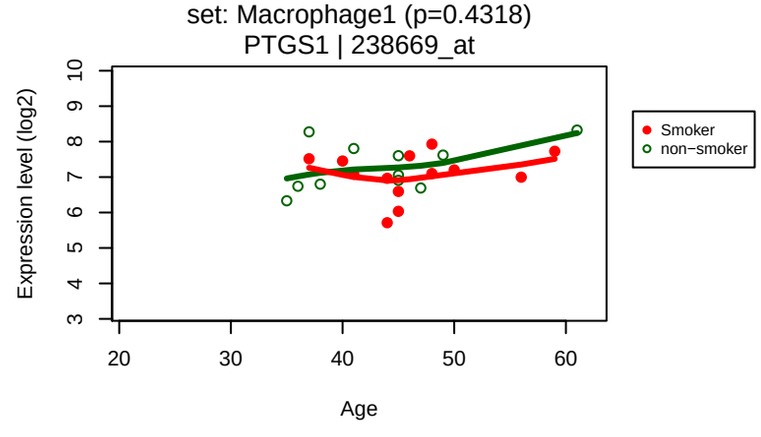

set: SmallAirway1 (p=0.5174)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker

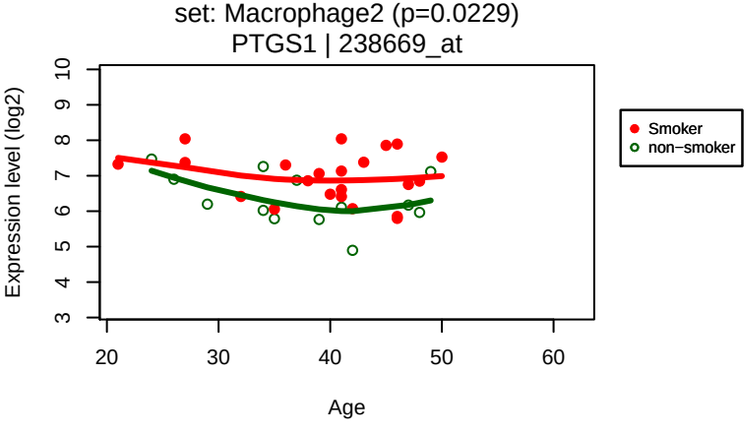

set: SmallAirway2 (p=0.2115)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker

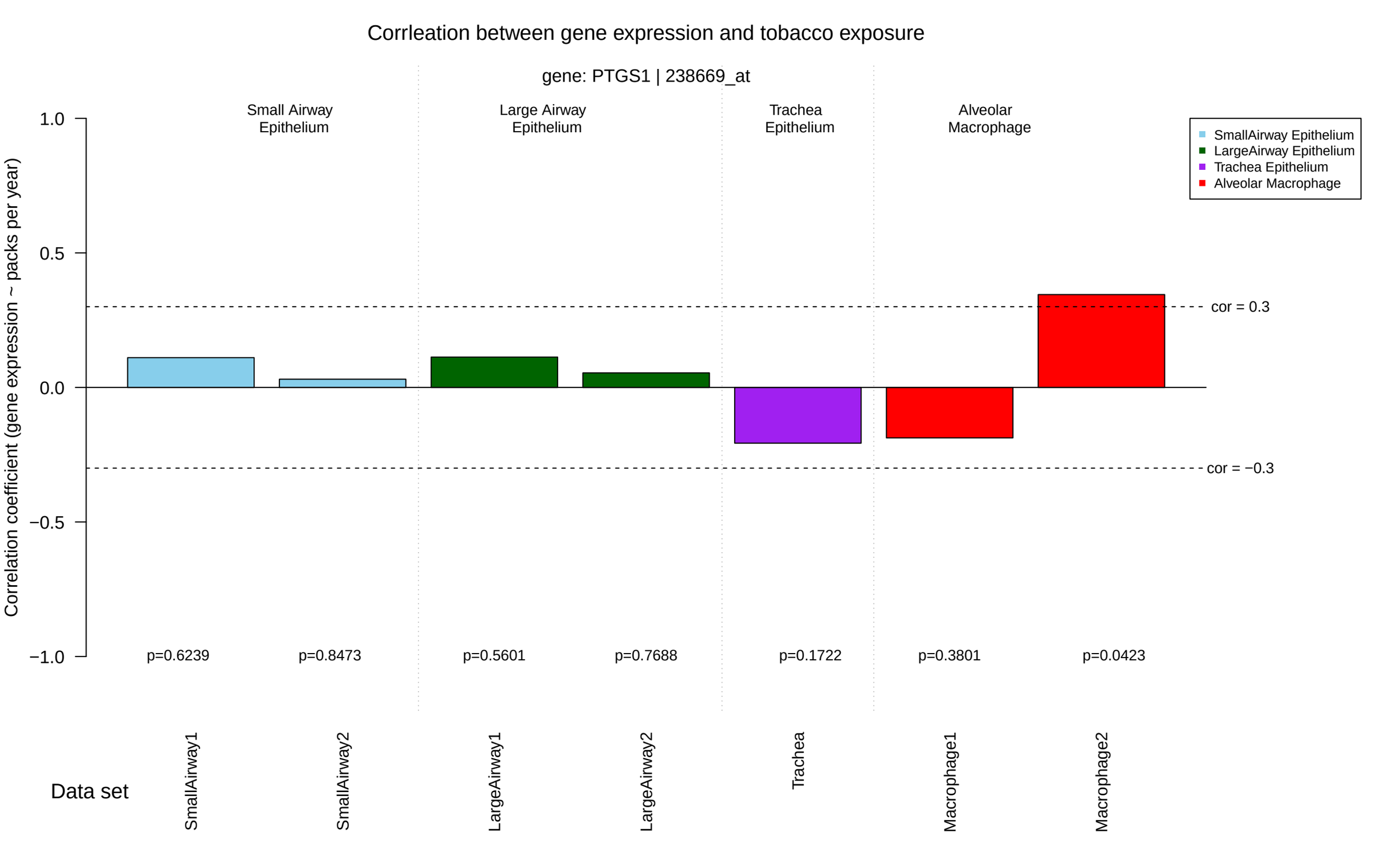

set: LargeAirway1 (p=0.2241)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.488)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0869)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4318)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0229)
PTGS1 | 238669_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGS1 | 238669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6239
p=0.8473
p=0.5601
p=0.7688
p=0.1722
p=0.3801
p=0.0423
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2